Lecture 10 recap

In Lecture 10:

- Introduced improper integrals and learnt how to assess their convergence
- Defined the Laplace transform and discussed its linear properties
- Set out criteria for existence of the Laplace transform of a function

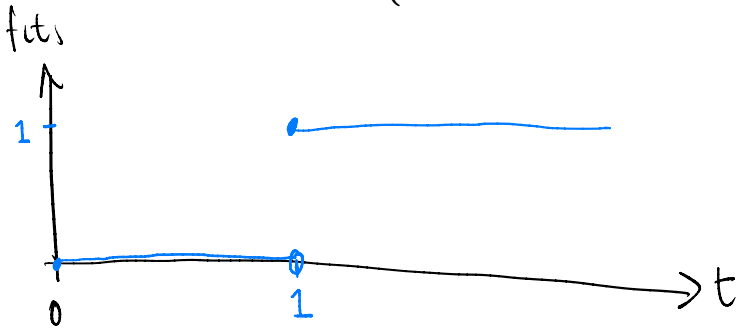
Lecture 11

In Lecture 11, we will:

1. Investigate existence of the Laplace transform for example functions
2. Introduce the inverse Laplace transform
3. Use the LT "cheat sheet" to calculate LT and ILT in examples

Exercise 3.6: Does the below function have a Laplace transform?

$$f = \begin{cases} 0 & 0 \leqslant t < 1 \\ 1 & 1 \leqslant t \end{cases}$$



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- Therefore, it is piecewise continuous
- It is exponentially bounded for large t
- We deduce that f has a Laplace transform

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Therefore, f_1 is bounded by an exponential and must have a Laplace transform.

f_2 grows faster than any exponential $k\,\mathrm{e}^{\alpha\,t}$ and so does not have a Laplace transform.

$\lim_{t\to\infty} e^{t^2}/e^{\alpha t} = \lim_{t\to\infty} \exp\{t(t-\alpha)\}$ but $t-\alpha > 0$ for $t > \alpha$ $\therefore$ lim DNE

Inverse Laplace transform

Theorem

- Let f be a continuous function on $t \in [0, \infty)$ that has a Laplace transform, F
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The inverse Laplace transform is linear:

1. $\mathscr{L}^{-1}\{F + G\} = \mathscr{L}^{-1}\{F\} + \mathscr{L}^{-1}\{G\} = f + g$
2. $\mathscr{L}^{-1}\{\beta F\} = \beta \mathscr{L}^{-1}\{F\} = \beta\, f$ where β is a constant

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2. If f is discontinuous at $\tau > 0$, the inverse returns the mid-point

$$\mathscr{L}^{-1}\{F\}\Big|_{t=\tau} = \frac{1}{2}\left(\lim_{t\to \tau^-} f(t) + \lim_{t\to \tau^+} f(t)\right)$$

MAST20030 Differential Equations Formulae Sheet

1) Laplace Transforms

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$ (Definition of Transform)
2.	1	$\frac{1}{s}$
3.	t^n	$\frac{n!}{s^{n+1}}$
4.	e^{at}	$\frac{1}{s-a}$
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

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Thus

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